In the above program, integer variables **num** and **avg** are declared and variable i is initialized to 1 and sum to 0. The while loop checks the condition, since i <= 5 the condition is true, a number is read from the user and this is added to sum and i is incremented by 1. Now, the condition is i <= 5 is again checked. Since 2 <=5, the second number is obtained from the user and it is added to sum. This continues, until i becomes 6, at which point the while loop terminates. After the loop termination, the avg is computed and both sum and avg are displayed.

While loop variation

A while loop may contain several variations. It can be an empty loop or an infinite loop. An empty while loop does not have any statement inside the body of the loop except null statement i.e., just a semicolon.

For example

```
int main()
{

    int i=0;
    while(++i < 10000 )
                          ------------>  This is an empty loop because the while
                                         loop does not contain any statement
    return 0;

}
```

In the above code, the loop is a time delay loop. A time delay loop is useful for pausing the program for some time.

A while loop may be infinite loop when no update statement is given inside the body of the loop. For example,

```
int main()
{

    int i = 0;
    while(i < =10)
      cout <<"The value of i is "<<i;  ------>  This statement will be displayed
                                                infinitely because no update
                                                statement inside the body of the loop
      i++;  ------------------------------>  This is not a part of the while loop statement
                                              because of missing curly braces
    return 0;


}
```


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